Biological Modelling Logistic Population Growth Model

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1 Question 1

Given the logistic population growth model;

1. Obtain the general solution to the model.
2. Determine the population size as $t \to \infty$.
3. Use python to simulate the solution to the model.

1.1 Problem statement

$$\frac{dN}{dt} = N(a - bN), \tag{1}$$

where:

- $N(t)$ is the population size at time t,
- $a > 0$ is the intrinsic growth rate of the population,
- $b > 0$ represents the effect of population density on growth.

The term aN represents population growth when the population is small, while the term bN^2 represents the reduction in growth caused by competition for limited resources.

The model can therefore be written as

$$\frac{dN}{dt} = aN - bN^2. \tag{2}$$

This is called the **logistic growth model** because it describes population growth that initially increases rapidly but eventually slows down as the population approaches a maximum sustainable level called the **carrying capacity**.

2 Part (i): Obtain the General Solution

We begin with

$$\frac{dN}{dt} = N(a - bN). \tag{3}$$

2.1 Step 1: Separate the Variables

We separate the variables N and t so that all terms involving N are on the left-hand side:

$$\frac{dN}{N(a - bN)} = dt. \tag{4}$$

This allows us to integrate both sides with respect to their respective variables.

2.2 Step 2: Partial Fraction Decomposition

We decompose

$$\frac{1}{N(a - bN)} \tag{5}$$

into simpler fractions:

$$\frac{1}{N(a - bN)} = \frac{A}{N} + \frac{B}{a - bN}. \tag{6}$$

Multiplying both sides by $N(a - bN)$ gives

$$1 = A(a - bN) + BN. \tag{7}$$

To determine A and B, we choose convenient values of N.
When $N = 0$,

$$1 = Aa, \tag{8}$$

so

$$A = \frac{1}{a}. \tag{9}$$

When $N = \frac{a}{b}$,

$$1 = B\left(\frac{a}{b}\right), \tag{10}$$

which gives

$$B = \frac{b}{a}. \tag{11}$$

Therefore,

$$\frac{1}{N(a-bN)} = \frac{1}{aN} + \frac{b}{a(a-bN)}. \tag{12}$$

Hence,

$$\frac{dN}{N(a-bN)} = \frac{1}{a}\left(\frac{1}{N} + \frac{b}{a-bN}\right) dN. \tag{13}$$

2.3 Step 3: Integrate

Integrating both sides gives

$$\int \frac{1}{a}\left(\frac{1}{N} + \frac{b}{a-bN}\right) dN = \int dt. \tag{14}$$

Therefore,

$$\frac{1}{a}\left[\ln|N| - \ln|a-bN|\right] = t + C_1. \tag{15}$$

The minus sign appears because

$$\int \frac{b}{a-bN}, dN = -\ln|a-bN|. \tag{16}$$

Multiplying by a gives

$$\ln|N| - \ln|a-bN| = at + aC_1. \tag{17}$$

Using the logarithm rule

$$\ln|x| - \ln|y| = \ln\left|\frac{x}{y}\right|, \tag{18}$$

we obtain

$$\ln\left|\frac{N}{a-bN}\right| = at + aC_1. \tag{19}$$

Exponentiating both sides,

$$\left|\frac{N}{a-bN}\right| = e^{aC_1}e^{at}. \tag{20}$$

The constant can be absorbed into a single arbitrary constant C, giving

$$\frac{N}{a - bN} = Ce^{at}. \tag{21}$$

2.4 Step 4: Solve for $N(t)$

We now solve the equation for N:

$$N = Ce^{at}(a - bN). \tag{22}$$

Expanding the right-hand side,

$$N = aCe^{at} - bCe^{at}N. \tag{23}$$

Collecting the terms containing N,

$$N + bCe^{at}N = aCe^{at}. \tag{24}$$

Factoring N,

$$N(1 + bCe^{at}) = aCe^{at}. \tag{25}$$

Thus,

$$N(t) = \frac{aCe^{at}}{1 + bCe^{at}}. \tag{26}$$

Dividing the numerator and denominator by Ce^{at} gives

$$N(t) = \frac{a}{b + \frac{1}{C}e^{-at}}. \tag{27}$$

2.5 Applying the Initial Condition

Suppose the initial population is

$$N(0) = N_0. \tag{28}$$

At $t = 0$,

$$N_0 = \frac{aC}{1 + bC}. \tag{29}$$

Multiplying through by $(1 + bC)$,

$$N_0 + bN_0C = aC. \tag{30}$$

Therefore,

$$N_0 = C(a - bN_0), \tag{31}$$

and hence

$$C = \frac{N_0}{a - bN_0}. \tag{32}$$

Substituting this into the solution gives

$$\boxed{N(t) = \frac{aN_0}{bN_0 + (a - bN_0)e^{-at}}}. \tag{33}$$

This is the logistic population growth solution.

2.6 Carrying Capacity Form

The carrying capacity is obtained by setting the growth rate equal to zero:

$$N(a - bN) = 0. \tag{34}$$

The non-zero equilibrium is

$$a - bN = 0, \tag{35}$$

so

$$K = \frac{a}{b}. \tag{36}$$

Therefore, the solution can also be written as

$$\boxed{N(t) = \frac{KN_0}{N_0 + (K - N_0)e^{-at}}}, \tag{37}$$

where

$$K = \frac{a}{b}. \tag{38}$$

This form is particularly useful for biological interpretation because K represents the maximum population size that the environment can sustainably support.

3 Part (ii): Determine the Population Size as $t \to \infty$

From the solution

$$N(t) = \frac{aN_0}{bN_0 + (a - bN_0)e^{-at}}, \tag{39}$$

assume that

$$a > 0. \tag{40}$$

As $t \to \infty$,

$$e^{-at} \to 0. \tag{41}$$

Therefore,

$$\lim_{t\to\infty} N(t) = \lim_{t\to\infty} \frac{aN_0}{bN_0 + (a - bN_0)e^{-at}} \qquad = \frac{aN_0}{bN_0} = \frac{a}{b}. \tag{42}$$

Since

$$K = \frac{a}{b}, \tag{43}$$

we conclude that

$$\boxed{\lim_{t\to\infty} N(t) = K = \frac{a}{b}}. \tag{44}$$

3.1 Biological Interpretation

The result means that the population approaches a stable maximum level rather than increasing indefinitely.

When the population is small, the term $a-bN$ is positive and the population increases.

As N becomes larger, the term bN becomes larger. This reduces the growth rate because of factors such as:

- Competition for food
- Competition for space
- Limited nutrients
- Predation
- Disease
- Other environmental limitations.

Eventually, the population approaches

$$K = \frac{a}{b}, \tag{45}$$

where the population growth becomes approximately zero.

Thus, the logistic model describes three general stages:

1. **Initial growth:** the population grows slowly when it is small.
2. **Rapid growth:** the population grows quickly when resources are relatively abundant.
3. **Stabilisation:** growth slows as the population approaches the carrying capacity.

4 Part (iii): Python Simulation

The analytical solution can be simulated using Python. Consider the parameters

$$a = 0.5, \qquad b = 0.005, \qquad N_0 = 10. \tag{46}$$

The carrying capacity is

$$K = \frac{a}{b} = \frac{0.5}{0.005}100. \tag{47}$$

Thus, the population is expected to approach 100 as time increases.

The following Python code calculates and plots the analytical solution.

```
import matplotlib.pyplot as plt
import numpy as np

# 1. Model Parameters
a = 0.5 # Growth rate
b = 0.005 # Parameter b
N0 = 10 # Starting population
K = a / b # Carrying capacity (100)

# 2. Time values and solution N(t)
t = np.linspace(0, 25, 100)
N = (a * N0) / (b * N0 + (a - b * N0) * np.exp(-a * t))

# 3. Numerical Summary
print(f"Starting␣Population␣(N0)␣:␣{N0}")
print(f"Carrying␣Capacity␣(a/b)␣␣:␣{K:.1f}")
```

```
print(f"Final␣Population␣(t=25)␣␣:␣{N[-1]:.2f}")
print(f"Average␣Population␣␣␣␣␣␣␣:␣{np.mean(N):.2f}")

# 4. Plot the population
plt.figure(figsize=(8, 5))
plt.plot(t, N, label="Population␣N(t)", color="blue", linewidth=2)
plt.axhline(
    y=K,
    color="red",
    linestyle="--",
    label=f"Carrying␣Capacity␣=␣{K:.0f}"
)

plt.title("Logistic␣Population␣Growth")
plt.xlabel("Time␣(t)")
plt.ylabel("Population␣Size␣(N)")
plt.legend()
plt.grid(True)

# Save the graph
plt.savefig("logistic_growth.png", dpi=300, bbox_inches="tight")

plt.show()
```

4.1 Output

The numerical output obtained from the simulation was:

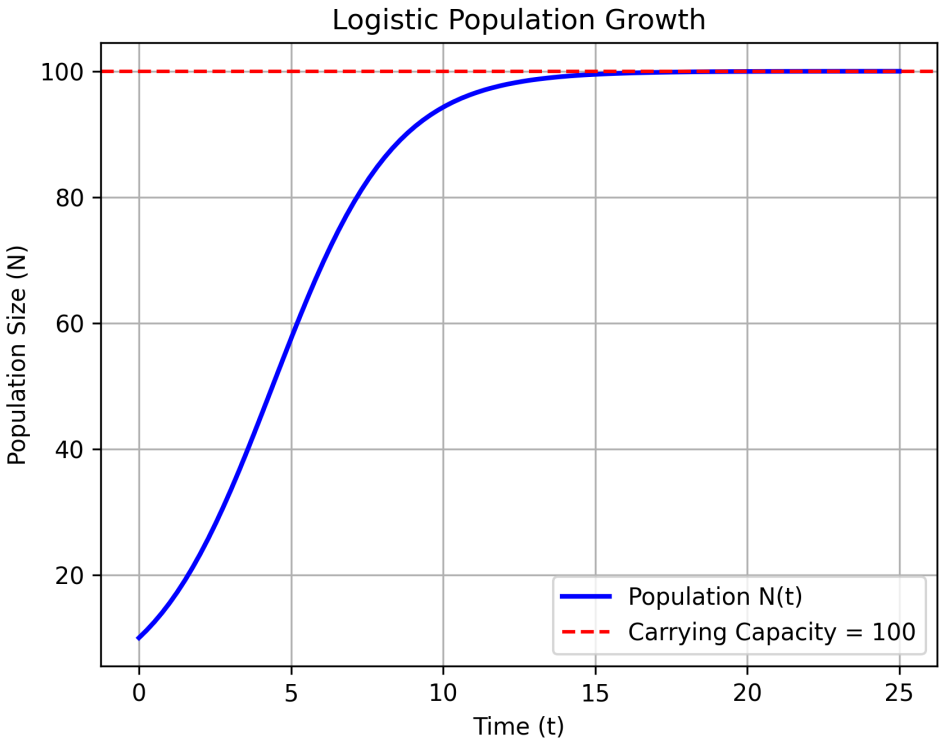


Figure 1: Simulation of logistic population growth.

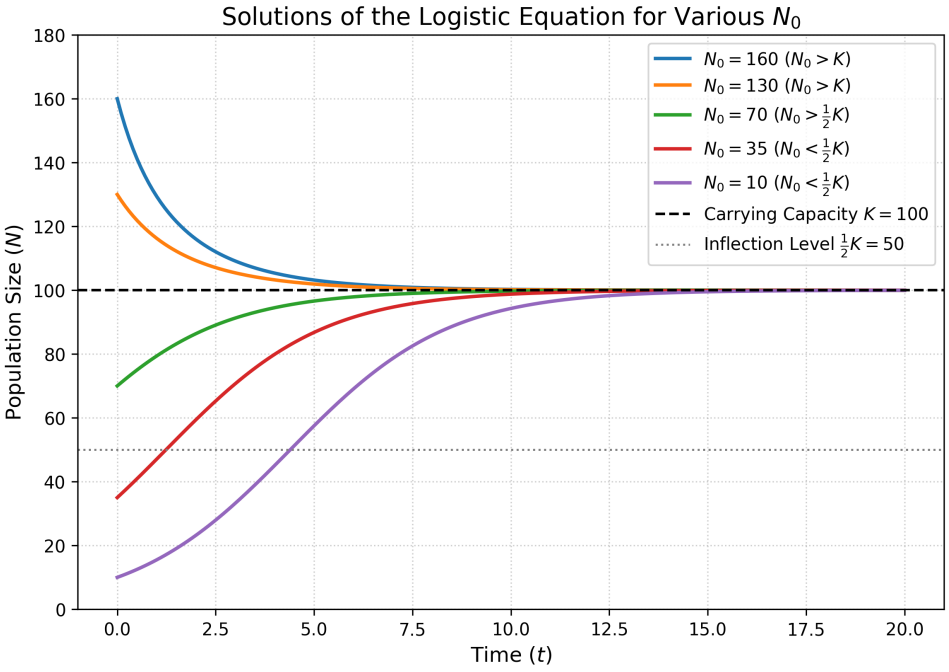


Figure 2: Simulation of logistic population growth.

4.2 Interpretation of the Simulation for multiple N0s

Figure 2 shows that the population increases from 10 and approaches the carrying capacity of 100.

For the chosen parameters,

$$N_0 = 10 \tag{48}$$

and

$$K = 100. \tag{49}$$

The graph shows the population starting at 10 and increasing towards 100.

Initially, the population grows rapidly because the population is well below the carrying capacity. As the population gets closer to 100, growth slows down. Eventually, the curve becomes almost horizontal around $N = 100$.

Therefore, the simulation confirms the mathematical result

$$\lim_{t\to\infty} N(t) = 100. \tag{50}$$

4.3 Explanation of Figure 2: Simulation of Logistic Population Growth

Figure 2 shows how a population $N(t)$ changes over time under the logistic growth model:

4.3.1 Critical Reference Lines

The graph contains two important reference levels.

- **Carrying Capacity ($K = 100$):** This is the population size that the environment can sustainably support. All population curves approach this value as $t \to \infty$: $\lim_{t\to\infty} N(t) = K = 100$.
- **Inflection Level ($K/2 = 50$):** This is the population size at which the growth rate is maximum. Below 50, the growth rate increases, while above 50, the growth rate decreases.

4.3.2 Different Population Behaviours

Case 1: Overpopulated ($N_0 > K$) The blue and orange curves represent $N_0 = 160$ and $N_0 = 130$, respectively. In both cases, the initial population is greater than the carrying capacity of 100.

The population decreases towards $K = 100$. The decrease occurs because the population is above the level that the environment can sustainably support.

Case 2: Moderately Sized Population ($K/2 < N_0 < K$) The green curve represents $N_0 = 70$. The population starts above the inflection level of 50 but below the carrying capacity of 100.

The population increases, but its growth rate gradually decreases as it approaches the carrying capacity.

Case 3: Small Starting Population ($N_0 < K/2$) The red and purple curves represent $N_0 = 35$ and $N_0 = 10$, respectively. These populations start below 50.

Initially, the growth rate increases as the population grows. When the population reaches $N = 50$, the growth rate is at its maximum. After passing 50, the growth rate decreases as the population approaches the carrying capacity of 100.

4.3.3 Summary of Population Behaviour

Table 1: Population behaviour under different initial conditions

Region	**Starting Condition**	**Growth Behaviour**	**Shap**
$N_0 > 100$	Decreases toward K	Concave up heightUpper Half	
Increases, but slows down	Concave down heightLower Half	$N_0 < 50$	Spe
S-shaped height			

4.3.4 Conclusion

The simulation shows that, for any positive initial population N_0, the population approaches the carrying capacity over time:

$$\boxed{\lim_{t\to\infty} N(t) = K = 100.}$$

Thus, although the populations start at different sizes, they eventually stabilise around the same carrying capacity.

5 Summary

The logistic model

$$\frac{dN}{dt} = N(a - bN) \tag{51}$$

describes population growth under environmental limitations.

The general solution with initial population $N(0) = N_0$ is

$$\boxed{N(t) = \frac{aN_0}{bN_0 + (a - bN_0)e^{-at}}}. \tag{52}$$

Alternatively, using the carrying capacity

$$K = \frac{a}{b}, \tag{53}$$

the solution is

$$\boxed{N(t) = \frac{KN_0}{N_0 + (K - N_0)e^{-at}}}. \tag{54}$$

As time approaches infinity,

$$\boxed{\lim_{t\to\infty} N(t) = \frac{a}{b} = K}. \tag{55}$$

Hence, the logistic model predicts that the population eventually stabilises at the carrying capacity rather than growing without limit.

6 Lotka-Volterra (Predator-Prey) System

The given Lotka-Volterra model is:

$$\frac{dx}{dt} = \alpha_1 x - \beta_1 xy \tag{56}$$

$$\frac{dy}{dt} = -\alpha_2 y + \beta_2 xy \tag{57}$$

6.1 (i) Identification of Predator and Prey

$x(t)$ **represents the Prey population:**

- In the absence of predators ($y = 0$),

$$\frac{dx}{dt} = \alpha_1 x > 0,$$

  meaning prey grows exponentially.

- The interaction term $-\beta_1 xy$ is negative, showing that encounters with y reduce the population of x.

$y(t)$ **represents the Predator population:**

- In the absence of prey ($x = 0$),

$$\frac{dy}{dt} = -\alpha_2 y < 0,$$

  meaning predators die out exponentially.

- The interaction term $+\beta_2 xy$ is positive, showing that encounters with x increase the population of y.

6.2 (ii) Description of the Parameters

- α_1**:** Intrinsic birth/growth rate of the prey in the absence of predators.
- β_1**:** Predation rate coefficient (the rate at which prey are consumed per encounter with predators).
- α_2**:** Natural mortality (death) rate of predators in the absence of prey.
- β_2**:** Predator reproduction efficiency (rate of converting consumed prey into new predator births).

6.3 (iii) Fixed Points (Equilibrium Points)

To find the fixed points, set the rates of change to zero:

$$\frac{dx}{dt} = x(\alpha_1 - \beta_1 y) = 0 \tag{58}$$

$$\frac{dy}{dt} = y(-\alpha_2 + \beta_2 x) = 0 \tag{59}$$

1. Extinction Equilibrium (Trivial Fixed Point):
If $x = 0$, then from the second equation:

$$y(-\alpha_2) = 0 \implies y = 0.$$

Therefore,

$$(x_1^*, y_1^*) = (0, 0) \tag{60}$$

2. Coexistence Equilibrium (Non-trivial Fixed Point):
If $x \neq 0$ and $y \neq 0$:

$$\alpha_1 - \beta_1 y = 0 \implies y^* = \frac{\alpha_1}{\beta_1}$$

$$-\alpha_2 + \beta_2 x = 0 \implies x^* = \frac{\alpha_2}{\beta_2}$$

Therefore,

$$(x_2^*, y_2^*) = \left(\frac{\alpha_2}{\beta_2}, \frac{\alpha_1}{\beta_1}\right) \tag{61}$$

6.4 (iv) Underlying Assumptions of the Model

1. In the absence of predators, prey has an ample food supply and grows exponentially without carrying capacity constraints.
2. The predator population relies solely on the prey species as its food source.
3. Encounter rates between predators and prey are directly proportional to the product of both populations ($x \cdot y$).
4. Spatial structure, seasons, and age distributions are ignored; individuals encounter each other randomly.
5. Births and deaths occur continuously at constant per-capita rates.

6.5 (v) Numerical Simulation and Explanation

6.5.1 Python Simulation Code (Euler's Method / ODE Solver)

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import odeint

# 1. Model Parameters
alpha_1 = 1.0 # Prey birth rate
beta_1 = 0.1 # Predation rate
alpha_2 = 1.5 # Predator death rate
beta_2 = 0.075 # Predator growth efficiency

# Equilibrium values
x_eq = alpha_2 / beta_2 # 20.0
y_eq = alpha_1 / beta_1 # 10.0

# 2. Differential Equations
def lotka_volterra(state, t):
    x, y = state
```

```
    dxdt = alpha_1 * x - beta_1 * x * y
    dydt = -alpha_2 * y + beta_2 * x * y
    return [dxdt, dydt]

# 3. Time Vector & Initial Conditions
t = np.linspace(0, 30, 1000)
initial_state = [30, 8] # [x0, y0]

# Solve numerically
solution = odeint(lotka_volterra, initial_state, t)
prey = solution[:, 0]
predator = solution[:, 1]

# 4. Visualization
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))

# (a) Time Series
ax1.plot(t, prey, "b-", label="Prey␣$x(t)$", linewidth=2)
ax1.plot(t, predator, "r--", label="Predator␣$y(t)$", linewidth=2)
ax1.set_title("Population␣Dynamics␣Over␣Time")
ax1.set_xlabel("Time␣($t$)")
ax1.set_ylabel("Population␣Size")
ax1.grid(True)
ax1.legend()

# (b) Phase Portrait
ax2.plot(prey, predator, "purple", linewidth=2)
ax2.scatter(
    [x_eq],
    [y_eq],
    color="black",
    zorder=5,
    label=f"Coexistence␣Point␣({x_eq:.0f},␣{y_eq:.0f})",
)
ax2.set_title("Phase␣Portrait␣($y$␣vs␣$x$)")
ax2.set_xlabel("Prey␣Population␣($x$)")
ax2.set_ylabel("Predator␣Population␣($y$)")
ax2.grid(True)
ax2.legend()

plt.tight_layout()
```

```
plt.show()
```

6.5.2 Results

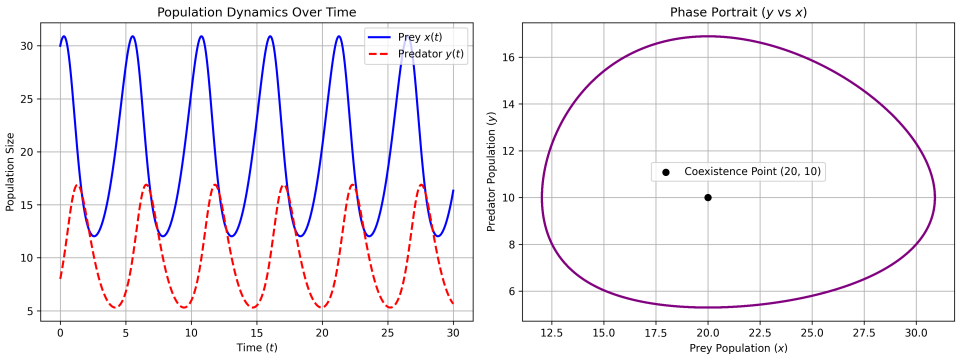


Figure 3: Simulation of Predator-Prey System.

6.5.3 Explanation of Results

1. **Periodic Oscillations (Cycles):** Both populations oscillate continuously in a repeating cycle. Neither population goes extinct or explodes to infinity.

2. **Phase Lag:** The prey population peaks first. As prey numbers rise, food abundance causes the predator population to grow, which in turn drives the prey population back down. Once prey becomes scarce, predators starve, allowing prey to recover, repeating the cycle.

3. **Closed Orbits in Phase Space:** In the phase plane (y vs. x), the trajectories form closed concentric loops around the non-trivial coexistence equilibrium

$$\left(\frac{\alpha_2}{\beta_2}, \frac{\alpha_1}{\beta_1}\right),$$

confirming neutral stability.